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# M-ALBA: A modelling framework to guide the optimization of membrane-assisted algae-bacteria systems

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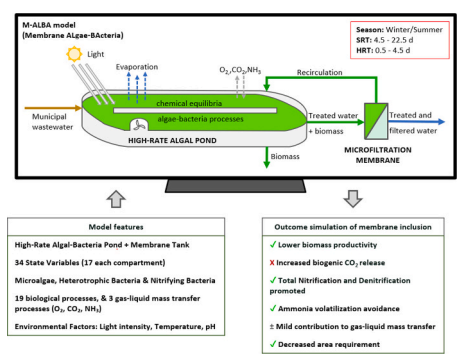
## HIGHLIGHTS

- The M-ALBA model represents membrane-algae-bacteria photobioreactors.
- Biomass retention increases biomass respiration rate, reducing algal productivity.
- High SRT promotes retention of nitrifying bacteria enabling total nitrification.
- Biomass retention increases treatment yield and reduces required area.
- Biomass retention enables operations at an HRT as low as 1.5 days.

## GRAPHICAL ABSTRACT

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## ABSTRACT

Biological systems combining microalgae and bacteria have been identified as a system with great potential to provide sustainable sanitation solutions. This consortium has been conceived to be normally implemented in the form of high-rate algal-bacteria ponds (HRABP). However, these systems face limitations, associated with effluent clarification and limited loads. Application of membrane filtration to induce biomass retention and effluent clarification have been identified as way to overcome such constraints. However, the effects of decoupling solid retention time (SRT) from hydraulic retention time (HRT) are complex and sometimes difficult to determine or predict. In this study, a model (M-ALBA) was used to predict the performance of a membrane-assisted HRABP. M-ALBA represents an extension of the previously validated ALBA model, by incorporating a compartment providing membrane separation. M-ALBA considers the action of microalgae, heterotrophic bacteria, and nitrifying bacteria (ammonium oxidizers and nitrite oxidizers), including 34 state variables, 19 biological processes and gas-liquid mass transfer of $O_2$, $CO_2$, and $NH_3$. Experimental data from previous study were used to evaluate the model accuracy. Different scenarios were simulated and analysed, using mass balances, considering SRT and HRT in the ranges 4.5–22.5 and 0.5–4.5 days, respectively. Results show how decoupling SRT from HRT improves effluent quality, by increasing nitrogen removal, while avoiding ammonia volatilization.

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Additionally, it allows operation at lower HRT values, achieving the best performance at HRT 1.5 days. The results obtained in this study contributed to a better understanding of the complex microalgae-bacteria dynamics in membrane-assisted HRABPs.

## Nomenclature

*Wastewater systems*

MBC microalgae-bacteria consortia
HRABP high-rate algal-bacteria ponds
MBR membrane bioreactor
MBR-MBC MBC in membrane bioreactor (MBR)
HRT hydraulic retention time
SRT solid retention time

*Modelling*

ALBA Algae-Bacteria model proposed by Casagli et al. (2021b)
M-ALBA ALBA model for uncoupling hydraulic and solid retention time
$X_i$ Particulate component *i*
$S_j$ Soluble component *j*
ALG microalgae
HB heterotrophic bacteria
NB nitrifying bacteria
AOB ammonium oxidizing bacteria
NOB nitrite-oxidizing bacteria

*System's flow configuration*

$Q_{in}$ influent flow of wastewater
$Q_w$ waste biomass flow exiting the system directly from the reactor
$Q_{out}$ reactor outflow entering the membrane tank
$Q_{ev}$ evaporation rate
$Q_p$ permeate flow
$Q_R$ retentate flow
$V_r$ reactor of constant volume
$V_m$ membrane tank volume
$K_La$ Volumetric oxygen mass transfer coefficient

*Model testing*

NRR nitrogen removal rate
BPR biomass production rate
RMSE root mean square error
$R^2$ Pearson coefficient of correlation

*Carbon balance*

C-ORG,P Particulate organic carbon (the sum of carbon in $X_S$ and $X_I$)
C-ORG,S Soluble organic carbon (the carbon content in $S_S$ and $S_I$)
C-ALG, C-HB and C-NB Carbon in algae, heterotrophic bacteria, and nitrifying bacteria, respectively
C-INORG Soluble inorganic carbon
C-$CO_2$r and C-$CO_2$m The inorganic carbon in the form of $CO_2$ gas, in the reactor and the membrane compartments, respectively

*Nitrogen balance*

N-ORG Nitrogen in organic matter other than biomass (i.e. $X_S$, $S_S$, $X_I$, and $S_I$) and soluble nitrogen ($S_{ND}$)
N-ALG, N-HB and N-NB Nitrogen in algae, heterotrophic bacteria, and nitrifying bacteria, respectively
N-TAN Total ammonia nitrogen, $S_{NH}$
N-$NO_2$ and N-$NO_3$ Nitrous nitrogen and nitric nitrogen, respectively
N-$NH_3$r and N-$NH_3$m Ammonia gas, in the reactor and the membrane compartments, respectively
N-$N_2$ Nitrogen gas

*Oxygen balance*

OUR_ALG, OUR_HB, OUR_NB oxygen uptake rate of microalgae, heterotrophs and nitrifiers, respectively
OPR_ALG oxygen production rate of algae from photooxygenation
OTRr and OTRm oxygen transfer rate from gas to liquid, in the reactor and the membrane compartments, respectively
CTRr and CTRm $CO_2$ transfer rate from gas to liquid, in the reactor and the membrane compartments, respectively
DO_AVE dissolved oxygen in the liquid phase

## 1. Introduction

Emerging biological wastewater treatment systems integrating microalgae and bacteria cultivation offers an eco-friendly alternative to reduce greenhouse gases emissions and energy consumption (Molazadeh et al., 2019; Vuppaladadiyam et al., 2018). Photosynthetic algae fix $CO_2$ while oxygen is produced, reducing the need for mechanical aeration and therefore the energy demand. Life cycle assessment studies have shown a reduction of 55 % in $CO_2$ emissions, when compared with conventional activated sludge systems (Arashiro et al., 2018; Garfí et al., 2017). Furthermore, the synergistic interactions facilitated by the co-cultivation of algae and bacteria foster nutrient removal efficiency, especially nitrogen and phosphorus, and facilitates organic matter degradation (Amini et al., 2020; Ji et al., 2018; Rezvani and Sarrafzadeh, 2023; Unnithan et al., 2014; Wang et al., 2015).

Microalgae-bacteria consortia (MBC) are often applied for sanitation in high-rate algal-bacteria ponds (HRABP), whose construction expenses are 70 % lower than conventional activated sludge systems (Nguyen et al., 2021; Young et al., 2017). Thus, algae-bacteria systems are progressively seen as an affordable and sustainable solution with the potential to target the future challenges of sanitation (Qiao et al., 2020). MBC for wastewater treatment is often operated in continuous mode, to achieve better treatment capabilities and nutrient removal efficiencies (Amini et al., 2020; Castellanos, 2013; Muñoz and Guieysse, 2006).

Microalgae-based systems still present drawbacks that need to be addressed, such as low volumetric capacity and low biomass settleability, which can affect effluent clarification (Olguín, 2012). Both problems can be addressed using membrane separation systems. Membrane systems have been reported as efficient tools for microalgae separation (Pavez et al., 2015; Saidane et al., 2021), providing an effluent free of suspended solids. Moreover, biomass retention can be used to increase conversion rate, improving volumetric capacity. Application of membrane bioreactor (MBR) concept to MBC systems (MBR-MBC) has been only recently explored (Arango et al., 2023; Robles et al., 2020; Sun et al., 2018). MBR technology promotes biomass retention by decoupling hydraulic from solid retention time (HRT and SRT) (Solmaz and Mustafa, 2019). As a result, SRT becomes a key parameter, shaping the dynamics of microbial species (Arango et al., 2023; Barbera et al., 2020; Rada-Ariza et al., 2019).

Determining the optimal value of SRT for wastewater treatment by MBR-MBC systems is not straightforward (Arango et al., 2022). Many factors can affect wastewater treatment such as wastewater characteristics and environmental conditions, including light intensity and

temperature. For instance, balancing biomass concentration to avoid self-shading while maximizing growth rate is complex (Foladori et al., 2020; Luo et al., 2017; Sepehri et al., 2020; Yang et al., 2018).

Mathematical modelling offers a systematic approach to understand and optimize operations and processes. As for other processes in wastewater treatment (Batstone et al., 2006; Henze et al., 2008), numerical tools can guide and optimize the design and operation of algae-bacteria systems. While several models exist for microalgal systems (Arango et al., 2022; Shoener et al., 2019), only a few works have addressed microalgae-bacteria systems (Broekhuizen et al., 2012; Nordio et al., 2024). Furthermore, the modelling of membrane-assisted MBC systems remains almost unexplored (Aparicio et al., 2023; Barbera et al., 2020). Recently, Casagli et al. (2021b) developed the ALBA model (ALgae-BActeria), designed to describe an MBC within a HRABP system. The ALBA model has been validated along three case studies, covering a large range of environmental and operational conditions (Narbonne, France; Milan, Italy; Almeria, Spain). These studies supported by experimental campaigns gathering >3 years of monitoring provide a strong confidence in the model for representing the biological dynamics. It accurately predicted daily and seasonal behaviours, including winter responses, which previous models failed to describe effectively. Casagli et al. (2023) later introduced a retention factor in the hydraulic balance for all particulate variables, to study the role of SRT and HRT in the optimization of microalgae productivity. However, the implementation of a dedicated membrane compartment is necessary to gain a deeper understanding of the system's behaviour and performance when a membrane-based approach is applied to separate SRT and HRT. Despite evident advances in the field, there are still great challenges when studying modelling of MBR-MBC systems. Some of them related with the effect of SRT and HRT on carbon and ammonia emissions, the impact of an aerated membrane tank on gases absorption or desorption, and the capacity of membrane retention to increase system capacity.

Here, we propose a model describing MBR-MBC systems, based on ALBA model. This model, named M-ALBA, involves the combination of a raceway reactor with an aerated membrane tank. Contribution of both compartments were considered for overall gases absorption and desorption. The M-ALBA model was then used to explore the effects of decoupling SRT from HRT on the system performance efficiency, focusing on nitrogen removal, ammonia volatilization, and biomass productivity, under different operational and environmental conditions. The generated data offer deeper insights into system performance and optimization and provide guideline for an efficient process operation. The modelling framework proposed in this study will be instrumental for environmental and economic evaluations and help in decision-making for large-scale applications.

## 2. Materials and methods

### 2.1. ALBA model overview

The ALBA model was developed by Casagli et al. (2021b) to evaluate wastewater treatment performance in a HRABP. The model includes 17 state variables, representing both particulate ($X_i$) and soluble ($S_j$) components (summarized in Table 1), 19 biological processes and 3 gas-liquid mass transfer processes. The biological consortium involves microalgae (ALG), heterotrophic bacteria (HB), and nitrifying bacteria (NB). NB include ammonium oxidizing bacteria (AOB) and nitrite-oxidizing bacteria (NOB).

In the ALBA model, microalgae grow using ammonium ($NH_4^+$) or nitrate ($NO_3^-$), phosphate, inorganic carbon in the form of carbon dioxide ($CO_2$) and bicarbonate ($HCO_3^-$), and light as an energy source. During growth, microalgae produce oxygen ($O_2$). HB grow on organic compounds, aerobically and consuming $NH_4^+$ over $NO_3^-$. However, when $O_2$ is unavailable, there is a shift to anoxic growth with both nitrite ($NO_2^-$) and $NO_3^-$ as electron acceptors. HB are also responsible for the hydrolysis of slowly biodegradable organic matter and urea (ammonification). As for nitrifiers, AOB convert ammonia ($NH_3$) into nitrite, while NOB complete nitrification process by oxidizing nitrite into nitrate. Nitrifiers grow aerobically, using inorganic carbon as a carbon source. For all the microbial populations, growth rate is determined by the limiting factor among multiple substrates (carbon, nitrogen, oxygen and phosphorus) following the Liebig's minimum law (de Baar, 1994).

The model includes respiration process for all types of microorganisms, resulting in biomass loss, subsequent oxygen consumption, and $CO_2$ release. Furthermore, decay processes are considered for all microorganisms, involving no oxygen consumption and resulting in the release of nutrients and organic matter.

Biological growth and respiration rates are temperature-dependent, as described by the Cardinal Temperature Model with Inflection (CTMI) proposed by Rosso et al. (1993), while decay rates follow the Arrhenius law. Additionally, pH dependence is incorporated using the Cardinal pH Model (CPM) proposed by Rosso et al. (1995) and includes a charge balance for all cationic and anionic chemical species. For microalgae, growth also depends on light, which is modelled using a Haldane-type expression. This function is integrated across the reactor's liquid depth, with light penetration calculated based on Lambert-Beer's law, accounting for both depth and algal biomass concentration.

In terms of physical processes, the model incorporates the

**Table 1**
State variables of the M-ALBA model.

| State variables | | Symbol | | Description | Unit |
|---|---|---|---|---|---|
| | | Reactor | **Membrane compartment** | | |
| Biomass | Particulate ($X$) | $X_{ALG,r}$ | $X_{ALG,m}$ | Phototrophic algae | g COD $m^{-3}$ |
| | | $X_{AOB,r}$ | $X_{AOB,m}$ | Ammonium Oxidizing Bacteria (AOB) | g COD $m^{-3}$ |
| | | $X_{NOB,r}$ | $X_{NOB,m}$ | Nitrite Oxidizing Bacteria (AOB) | g COD $m^{-3}$ |
| | | $X_{H,r}$ | $X_{H,m}$ | Heterotrophic Bacteria (HB) | g COD $m^{-3}$ |
| Substrate and products | | $X_{S,r}$ | $X_{S,m}$ | Slowly biodegradable organic matter | g COD $m^{-3}$ |
| | | $X_{I,r}$ | $X_{I,m}$ | Inert particulate organic matter | g COD $m^{-3}$ |
| | Soluble ($S$) | $S_{S,r}$ | $S_{S,m}$ | Readily biodegradable organic matter | g COD $m^{-3}$ |
| | | $S_{I,r}$ | $S_{I,m}$ | Inert soluble organic matter | g COD $m^{-3}$ |
| | | $S_{IC,r}$ | $S_{IC,m}$ | Total inorganic carbon | g C $m^{-3}$ |
| | | $S_{ND,r}$ | $S_{ND,m}$ | Organic nitrogen | g N $m^{-3}$ |
| | | $S_{NH,r}$ | $S_{NH,m}$ | Total Ammoniacal Nitrogen (TAN) | g N $m^{-3}$ |
| | | $S_{NO2,r}$ | $S_{NO2,m}$ | Nitrous nitrogen | g N $m^{-3}$ |
| | | $S_{NO3,r}$ | $S_{NO3,m}$ | Nitric nitrogen | g N $m^{-3}$ |
| | | $S_{N2,r}$ | $S_{N2,m}$ | Nitrogen gas | g N $m^{-3}$ |
| | | $S_{PO4,r}$ | $S_{PO4,m}$ | Total inorganic phosphorus | g P $m^{-3}$ |
| | | $S_{O2,r}$ | $S_{O2,m}$ | Dissolved Oxygen | g $O_2$ $m^{-3}$ |
| | | $S_{H2O,r}$ | $S_{H2O,m}$ | Water | g H $m^{-3}$ |

desorption/absorption of $CO_2$, $NH_3$, and $O_2$ between atmosphere and liquid phase. Saturation values for these processes vary with temperature according to Henry's law.

A sensitivity analysis under periodic regime across different seasons revealed the most sensitive model parameters, which were subsequently calibrated. The model has been validated through three case studies, representing a broad range of environmental and operational conditions (Narbonne, France; Milan, Italy; Almeria, Spain). These studies, supported by more than three years of monitoring data from experimental campaigns, provide strong evidence of the model's reliability. Additionally, an uncertainty analysis was conducted to quantify the prediction errors, accounting both from measurement and parameters inaccuracies. Further details on the procedures followed for sensitivity analysis, calibration, validation, and uncertainty analysis, according to the different case studies, can be found in literature (Casagli et al., 2021b; Casagli et al., 2021a; Casagli and Bernard, 2022).

### 2.2. M-ALBA model

The M-ALBA model is an extension of the ALBA model, including a dedicated membrane compartment for the uncoupling of SRT from HRT. Then, M-ALBA considers a two-compartment system: a HRABP and a membrane tank, with a flow between the two systems (Fig. 1). M-ALBA involves 17 state variables for each compartment, 34 in total (see Table 1). The reactor compartment is characterized by 19 biological processes and the gas/liquid transfers of $O_2$, $CO_2$, and $NH_3$, as is the case of ALBA model. Biological activity within the membrane compartment is neglected, only occurring gas/liquid mass transfer of $O_2$, $CO_2$, and $NH_3$. This hypothesis is justified by the considerably lower HRT of the membrane compartment compared to that of the reactor. It is worth noting that the M-ALBA model exactly coincides with the ALBA model when SRT equals HRT. As Fig. 1 indicates, the filter module is located at the exit of the reactor, as is commonly used for MBRs.

The influent wastewater flow ($Q_{in}$) enters the reactor of constant volume $V_r$ where it is treated. A biomass outflow ($Q_w$) directly leaves the reactor, while another outflow ($Q_{out}$) enters the membrane compartment of volume $V_m$. The water mass flow resulting from water evaporation is considered ($Q_{ev}$). Inside the membrane tank, the membrane separation provides complete retention of suspended solids. Then, a permeate flow leaves the membrane compartment ($Q_p$) and the retentate flow ($Q_R$) is recycled to the reactor. All flow rates can be calculated as a function of applied HRT and SRT, according to Eqs. 1 to 5.

$$Q_{in} = \frac{V_r}{HRT} \tag{1}$$

$$Q_w = \frac{V_r}{SRT} \tag{2}$$

$$Q_p = Q_{in} - Q_{ev} - Q_w \tag{3}$$

$$Q_R = RQ_{in} \tag{4}$$

$$Q_{out} = Q_p + Q_R \tag{5}$$

$Q_R$ is assumed to be proportional to $Q_{in}$, where the proportionality constant R is chosen to ensure that particulate concentrations inside the membrane compartment do not exceed 10 % of that inside the MBR, to prevent excessive membrane fouling. Eqs. (3) and (5) result from constant compartments volumes.

The evolution of the state variables in the reactor of the particulate ($X_{j,r}$) and soluble ($S_{j,r}$) species are described by the following expressions:

$$\frac{dX_{j,r}}{dt} = \frac{Q_{in}}{V_r}X_{j,in} + \frac{Q_R}{V_r}X_{j,m} - \frac{Q_{out}}{V_r}X_{j,r} - \frac{Q_w}{V_r}X_{j,r} \pm \sum_i k_{i,j}\rho_i \tag{6}$$

$$\frac{dS_{j,r}}{dt} = \frac{Q_{in}}{V_r}S_{j,in} + \frac{Q_R}{V_r}S_{j,m} - \frac{Q_{out}}{V_r}S_{j,r} - \frac{Q_w}{V_r}S_{j,r} \pm \sum_i k_{i,j}\rho_i \pm Q_j \tag{7}$$

In the same way, membrane compartment dynamics for particulate ($X_{j,m}$) and soluble ($S_{j,m}$) components are:

$$\frac{dX_{j,m}}{dt} = \frac{Q_{out}}{V_m}X_{j,r} - \frac{Q_R}{V_m}X_{j,m} \tag{8}$$

$$\frac{dS_{j,m}}{dt} = \frac{Q_{out}}{V_m}S_{j,r} - \frac{(Q_R + Q_p)}{V_m}S_{j,m} \pm Q_j \tag{9}$$

where the term $\rho_i$ represents the biological reaction rate [g m$^{-3}$ d$^{-1}$] and $k_{i,j}$ is the stoichiometric coefficients associated to the state variable j. On the other hand, $Q_j$ denotes the gas/liquid transfer rate of $O_2$, $CO_2$, and $NH_3$.

### 2.3. Simulating large-scale MBR-MBC systems

The M-ALBA model was used to predict the performance of full-scale MBR-MBC, in order to gain insights of the large-scale mechanisms responsible for organic matter and nitrogen removals and eventually to propose guidelines for managing such system. A 5000 m$^2$ HRABP was considered, with depth 0.3 m meaning a volume of 1500 m$^3$. It was assumed that an appropriate mixing system such as paddle wheels ensures perfect reactor homogeneity. Immediately after the reactor, a membrane filtration compartment is positioned. Based on the recommendations for the ZeeWeed hollow fibre membranes, the membrane compartment volume was adapted for each HRT as a function of the required membrane area, assuming a constant permeate flux of 20 [L m$^{-2}$ h$^{-1}$]. This adjustment resulted in membrane volumes ranging from 20 to 68 [m$^3$]. For clarity, it is important to note that the system HRT is defined in terms of reactor volume (see Eq. 1), excluding the membrane compartment volume.

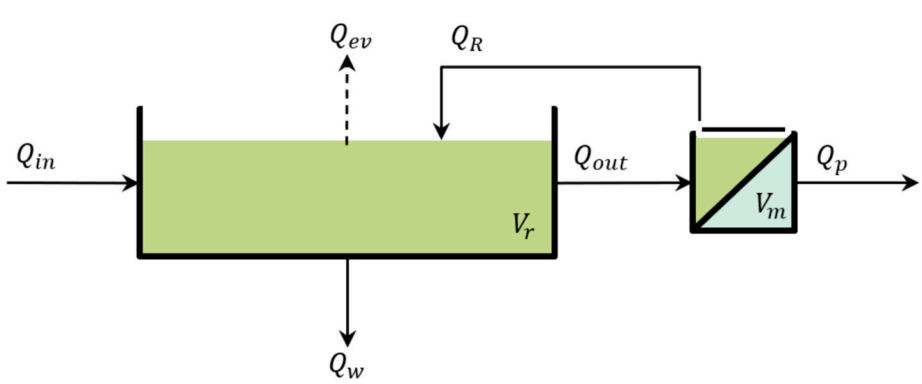


**Fig. 1.** Schematic diagram of the studied MBR-MBC. The volume of the reactor and the membrane compartment are denoted as $V_r$ and $V_m$, respectively. $Q_{in}$, $Q_{ev}$, $Q_w$, $Q_{out}$, $Q_p$ and $Q_R$ represent the flows associated with influent, evaporation, purge, effluent, permeate and recirculation, respectively.

Different operational conditions were simulated, considering various SRT (4.5, 9, 13.5, 18 and 22.5 days) and HRT values (0.5, 1.5, 2.5, 3.5, and 4.5 days). Operations involving equal SRT and HRT were simulated using the standard ALBA model, i.e., without a membrane compartment. Two different climatological conditions were considered, representing typical conditions for winter and summer in the south of France (Narbonne). Each climatological condition was defined by a periodic daily pattern of light, temperature, and evaporation rate (see Fig. 2). The corresponding average, maximum and minimum value of each parameter are detailed in Table 2. The wastewater characteristics of the influent were the same as in Casagli et al. (2021b) (see Table 2).

The simulations were run using MATLAB, specifically using the ode15s solver, which is well-suited for addressing the stiffness arising from the coupled biological and chemical-physical processes in the M-ALBA model (Shampine and Reichelt, 1997).

Global volumetric mass transfer coefficient $K_La$ was set at 20 [$d^{-1}$] for the reactor (El Ouarghi et al., 2003). A conservative value of 200 [$d^{-1}$] inside the membrane tank was considered (Germain et al., 2007; Henkel et al., 2009), resulting from the intense gas sparging normally used to prevent membrane fouling in hollow-fibre membranes.

Finally, for each operational and climatological condition, the M-ALBA model was run until a periodic regime on the state variables was reached. Average estimations of biomass productivity, oxygen and $CO_2$ fluxes were then assessed, together with carbon and nitrogen balances.

### 2.4. M-ALBA model validation

The biological kinetics in the M-ALBA model and parameters are identical to the one established and validated for the ALBA model, which demonstrated their efficiency to reproduce a large range of operational modes and environmental conditions on the long run.

The additional experimental data used for M-ALBA model validation, when SRT and HRT differ, were obtained from an experimental campaign conducted by Robles et al. (2020) in the south of France (Narbonne), using the same outdoor HRABP of 16.8 $m^3$ (56 $m^2$), on which the original ALBA model was calibrated and validated. These data include nitrogen removal rate (NRR), biomass production rate (BPR), irradiance, and temperature. The reactor was coupled to external microfiltration membrane and fed with synthetic municipal wastewater, similar to that used in the study by Casagli et al. (2021b). Experiments spanning about 12 weeks were considered, during which high-frequency data on key process variables were collected. These datasets captured the dynamic behaviour of the system under real outdoor conditions, submitted to permanent fluctuations of light and temperature conditions. NRR and BPR were evaluated under three different operational conditions, all with a fixed SRT of 6 days, but varying HRTs: (i) HRT = 2.5 days, during autumn; (ii) HRT = 4 days, during autumn; and (iii) HRT = 2.5 days, during summer. More specific details about the experimental campaign can be found in Robles et al. (2020).

Following the work of Casagli et al. (2023) the NRR and BPR were simulated using M-ALBA and then compared to the experimental values using the root mean square error (RMSE) and the Pearson correlation coefficient ($R^2$). The current validation focused on BPR and NRR is intended to demonstrate that the model can predict key wastewater treatment parameters accurately, under different operational conditions. Once the M-ALBA model has been validated, showing to accurately represent reality, it was used to explore hypothetical scenarios, with different combinations of SRT and HRT.

## 3. Results and discussions

### 3.1. M-ALBA accuracy on nitrogen removal and biomass productivity

Figs. 3.A and 3.B present NRR and BPR, with a comparison between the experimental data by Robles et al. (2020) and the M-ALBA model predictions. Overall, the M-ALBA showed a good performance for predicting the BPR and NRR. Average simulated values of BPR ranged from 16.8 to 22.8 [g VSS $m^{-2}$ $d^{-1}$], showing a small model underestimation (below 20 %). Analysis revealed an RMSE of 3.7 [g VSS $m^{-2}$ $d^{-1}$] and $R^2$ coefficient equal to 0.90. On the other hand, the average NRR ranging from 2.0 to 3.4 [g $N^{-1}$ $m^{-2}$ $d^{-1}$] with no clear trend. RMSE for nitrogen removal was 0.3 [g $N^{-1}$ $m^{-2}$ $d^{-1}$] whereas $R^2$ was 0.71. Predicting more accurately the system behaviour is challenging due to the inherent operative difficulties expected for an open pilot plant installation. Also, the initial conditions, in terms of the various populations initially present were not measured. Despite all these uncertainties, the model predictions well predict the experimental records.

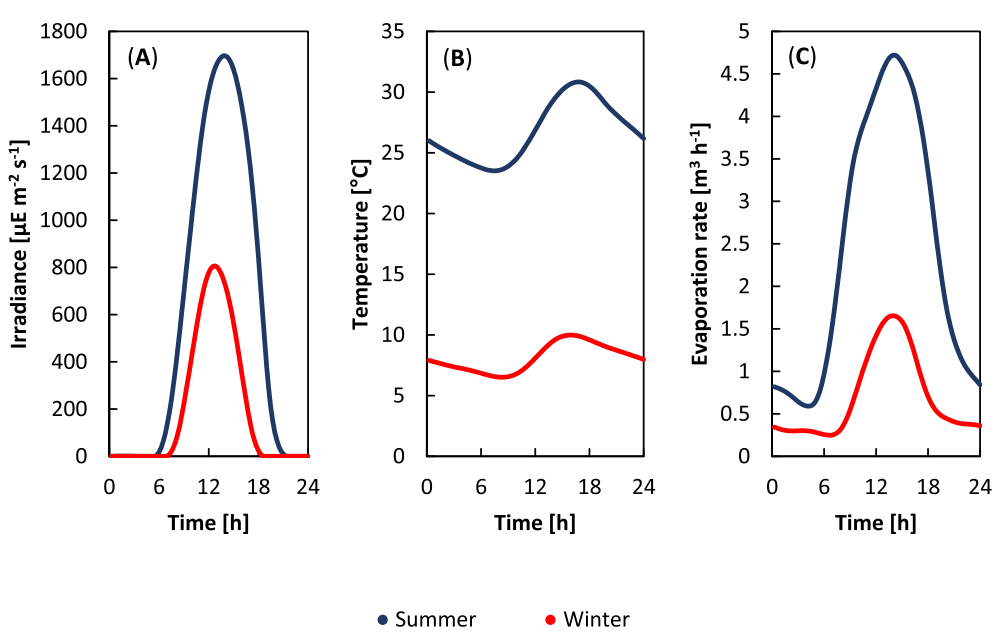


**Fig. 2.** Typical average daily patterns of (A) Incident photosynthetically active radiation (PAR), (B) Temperature, (C) Evaporation in summer (blue line) and winter (red line) (Casagli et al., 2021b).

**Table 2**
Influent values of the state variables included in the M-ALBA model and environmental conditions (Casagli et al., 2021b).

| Influent characteristics | | | | | | | |
|---|---|---|---|---|---|---|---|
| | $X_{ALG}; X_H; X_{AOB}; X_{NOB}$ | $X_S$ | $X_I$ | $S_S$ | $S_I$ | $S_{IC}$ | $S_{NH}$ |
| Unit | g COD $m^{-3}$ | g COD $m^{-3}$ | g COD $m^{-3}$ | g COD $m^{-3}$ | g COD $m^{-3}$ | g C $m^{-3}$ | g N $m^{-3}$ |
| Value | 0 | 133 | 3 | 224 | 18 | 65 | 8 |
| | $S_{NO2}$ | $S_{NO3}$ | $S_{N2}$ | $S_{ND}$ | $S_{PO4}$ | $S_{O2}$ | $S_{H2O}$ |
| Unit | g N $m^{-3}$ | g N $m^{-3}$ | g N $m^{-3}$ | g N $m^{-3}$ | g P $m^{-3}$ | g $O_2$ $m^{-3}$ | g H $m^{-3}$ |
| Value | 0.03 | 0.05 | 0 | 55 | 13 | 8 | 10,000 |

| Environmental conditions (average; maximum) | | |
|---|---|---|
| | Summer | Winter |
| Incident irradiance [μE $m^{-2}$ $s^{-1}$] | 584; 1697 | 196; 807 |
| Temperature [°C] | 26.8; 30.8 | 8.1; 10.0 |
| Evaporation rate [$m^3$ $m^2$ $h^{-1}$] | 2.40; 4.77 | 0.69; 1.65 |

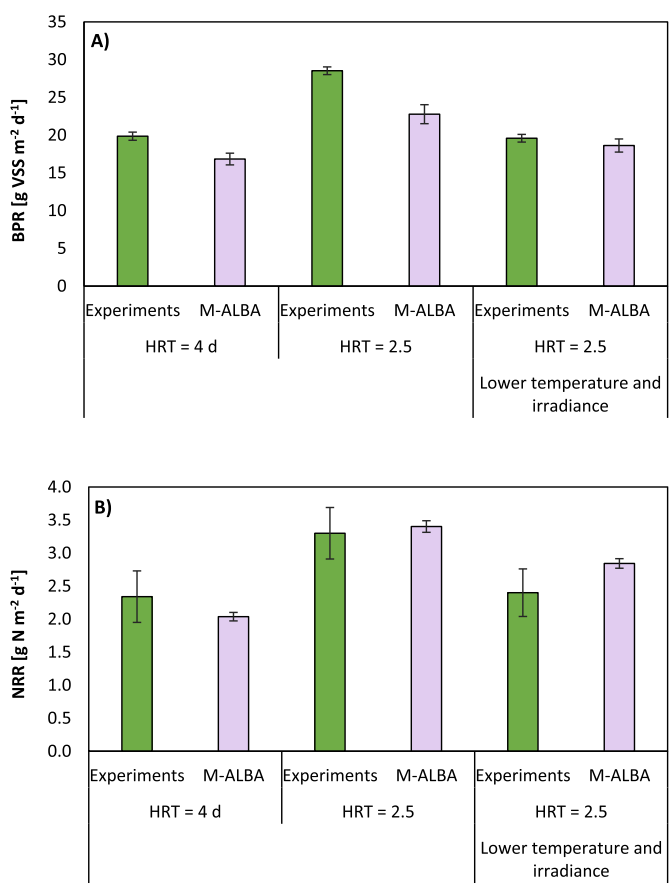


**Fig. 3.** Comparison between experimental data (Robles et al., 2020) and simulations with the M-ALBA model. A) Biomass productivity in [g VSS $m^{-2}$ $d^{-1}$], B) Nitrogen removal rate [g N $m^{-2}$ $d^{-1}$].

### 3.2. Behaviour of a large-scale MBR-MBC system

#### 3.2.1. Biomass composition

The simulated biomass compositions are presented in Fig. 4, together with the average biomass productivities (Px). We observe that the concentration of algae was mainly affected by SRT, while the concentration of HB was mainly affected by the HRT. Uncoupling SRT from HRT seems an efficient strategy to prevent the washout of the nitrifying biomass in the MBC for HRT larger than 0.5 [d]. Fig. 4 also shows that increasing SRT from 4.5 to 22.5 [d] almost doubled the total biomass concentration in all scenarios, due to the enhancement of the biomass retention. On the other hand, a reduction of HRT from 4.5 to 1.5 [d] triggered a total biomass concentration raise from 36 to 62 % for winter scenarios, and from 9 to 17 %, for summer scenarios. This is the direct consequence of the higher nutrient loading rate. A further reduction of HRT from 1.5 down to 0.5 [d] has even a more marked effect, with a 1.6 and 2.2-fold increase, for summer and winter respectively.

Fig. 4 shows that in most scenarios, especially during summer, microalgae are expected to dominate the microbial populations, followed by heterotrophs. Only during winter and at specific conditions, HB became dominant. For HRT of 1.5 [d] and SRT larger than 4.5 [d], the concentrations of HB and ALG were very similar, whatever the SRT applied. In summer, HB concentrations were approximately 42 % to 65 % lower than for winter, whereas microalgae concentrations were around 1.5 to 2 times higher. Seasonal differences in predicted HB concentrations are mainly attributed to temperature, since the model predicts that the impact of this parameter on decay rates is comparatively higher than its influence on growth rates. In addition, model predicts that higher temperature scenarios (summer) affect negatively the ratio between aerobic and anoxic growth rates. Finally, simulations analysis revealed that the limiting factor for microalgae growth was irradiance, and not nutrients as for heterotrophic organisms.

Fig. 4 also predicts that an increase in SRT results in a rise of biomass concentration but a decrease in biomass productivity. In fact, uncoupling SRT from HRT promotes higher biomass concentrations in the system, resulting in higher respiration rates. This shifts the fate of incoming carbon from contributing to biomass production towards generating biogenic emissions of $CO_2$, as addressed in the following section. For example, during the summer and at HRT = 1.5 [d], productivity decreased from 26.4 to 8.0 [g TSS $m^{-2}$ $d^{-1}$], when increasing SRT from 1.5 [d] to 22.5 [d]. On the other hand, decreasing HRT resulted in higher overall biomass productivity at any tested SRT (except at 0.5 [d]), mainly because of the growth of heterotrophs.

#### 3.2.2. Carbon mass balances

Carbon mass balances were evaluated for each scenario, and the results are presented in Fig. 5, identifying the distribution (in %) of each carbon flow. To simplify the representation of carbon flows, some model components were grouped as follows:

- C-ORG,P: Particulate organic carbon (the sum of carbon in $X_S$ and $X_I$).
- C-ORG,S: Soluble organic carbon (the carbon content in $S_S$ and $S_I$).
- C-ALG: Carbon in algae; C-NB: Carbon in nitrifying bacteria; and C-HB: Carbon in heterotrophic bacteria.
- C-INORG: Soluble inorganic carbon.
- C-$CO_2$r and C-$CO_2$m: The inorganic carbon stripped from the system in the form of $CO_2$ gas, from the reactor and the membrane compartment, respectively.

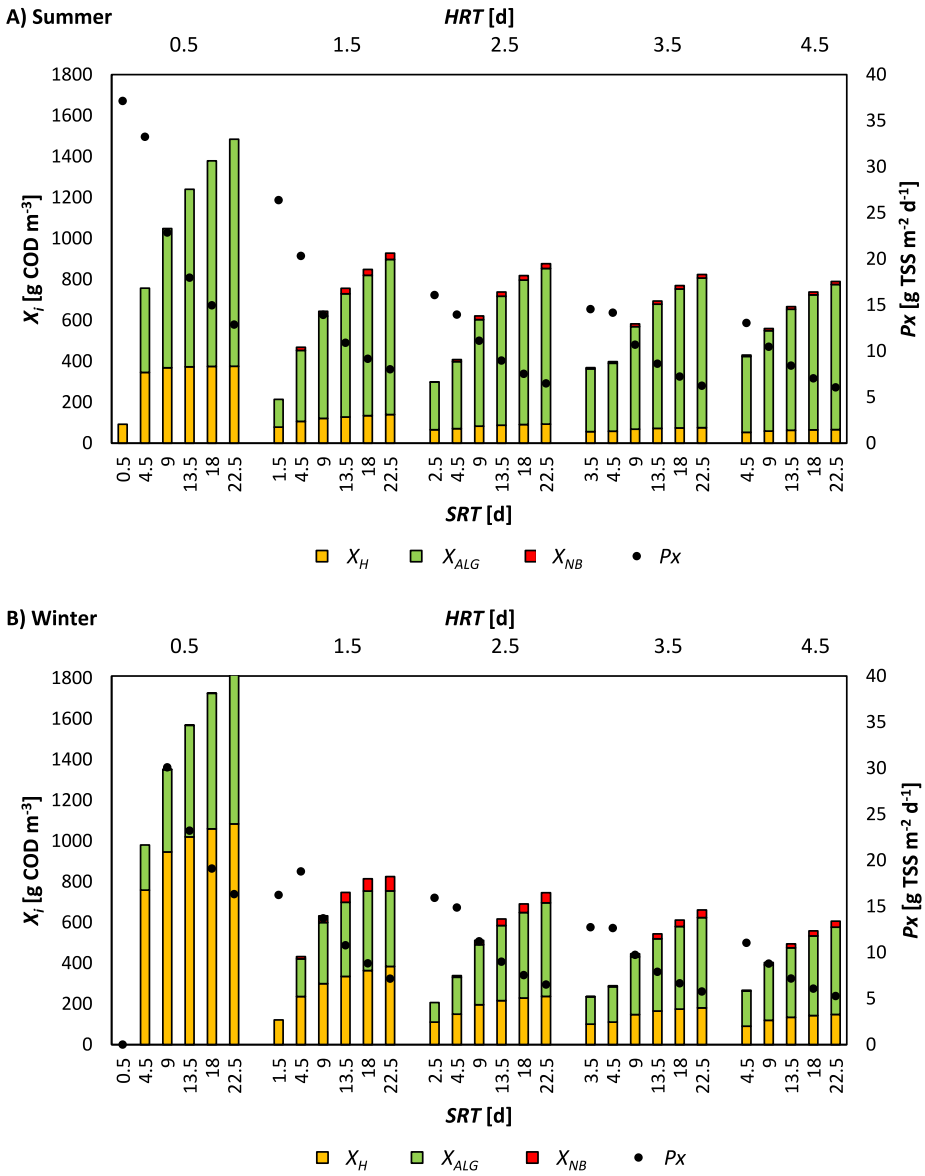


**Fig. 4.** Average concentration and productivity of biomass during summer (A) and winter (B) at different conditions of HRT and SRT. $X_H$ is the concentration of the heterotrophic bacteria, $X_{ALG}$ is the microalgae concentration, and $X_{NB}$ is the concentration of nitrifying bacteria. In the secondary axis of the graph, the average biomass productivity *Px* is shown with black dots.

Fig. 5 shows that biomass-associated carbon flows (C-ALG, C-NB and C-HB) presented the most important variations among the evaluated scenarios. These relative contributions were reduced when increasing the SRT, following the trends of biomass productivity as shown in Fig. 4.

At HRT values between 1.5 and 4.5 [d], and SRT > HRT, the sum of organic particulate (C-ORG,P) and organic soluble carbon (C-ORG,S) stayed in the range 18–22 %. In those scenarios, heterotrophic bacteria consumed almost entirely the biodegradable organic matter ($S_S \sim 0$ [g m$^{-3}$]). Hence C-ORG,S was mainly associated to inert soluble carbon ($S_I$). In contrast, in scenarios where HRT was set at 0.5 [d] the consumption of biodegradable organic matter was incomplete, so the C-ORG,S were much higher. Regarding C-ORG,P, the values did not undergo significant variations with respect to SRT, even though the concentration of inert organic particulates ($X_I$) increased significantly, because the wasted solids flow ($Q_w$) compensated the variation. For instance, for the summer scenario, the concentration of $X_I$ for SRT = 22.5 [d] at HRT = 1.5 [d] was 6-fold the value at SRT = 4.5 [d] (514.5 and 81.5 [g m$^{-3}$], respectively), but the mass carbon flux of the former was only 1.3 times of the latter (12.4 and 9.8 [kg C d$^{-1}$], respectively).

Fig. 5 also shows how the contribution of C-$CO_2$r and C-$CO_2$m to overall carbon balance increases as SRT increases, for the HRT scenarios between 1.5 and 4.5 [d]. This is the result of the reduction of the systems capability to assimilate inorganic carbon, associated with a lower biomass productivity. For example, at HRT = 1.5 [d] during summer, increasing the SRT from 4.5 to 22.5 [d] raised the sum of C-$CO_2$r and C-$CO_2$m from 32.0 to 42.0 %. It is important to emphasize that Fig. 5 represents contributions rather than actual carbon flows. Variations in HRT alter the influent flow rate, leading to significant changes in the absolute values of both influent and effluent carbon flows. The proportion of $CO_2$ emitted from the membrane compartment, C-$CO_2$m, increased when HRT decreased. Indeed, a higher inflow requires a larger membrane area, associated with a larger membrane compartment, and as a result a larger $CO_2$ stripping. Nevertheless, the contribution of the membrane tank to the overall $CO_2$ stripping stays relatively low for the tested working conditions. More precisely, the relation between the $CO_2$ transfer rates (absorption or desorption) in the reactor (CTRr) and the membrane tank (CTRm) depends on the $K_La$ values, the compartment volumes ($V_r$ and $V_m$) the $CO_2$ concentrations ($S_{CO2,R}$ and $S_{CO2,m}$) and the

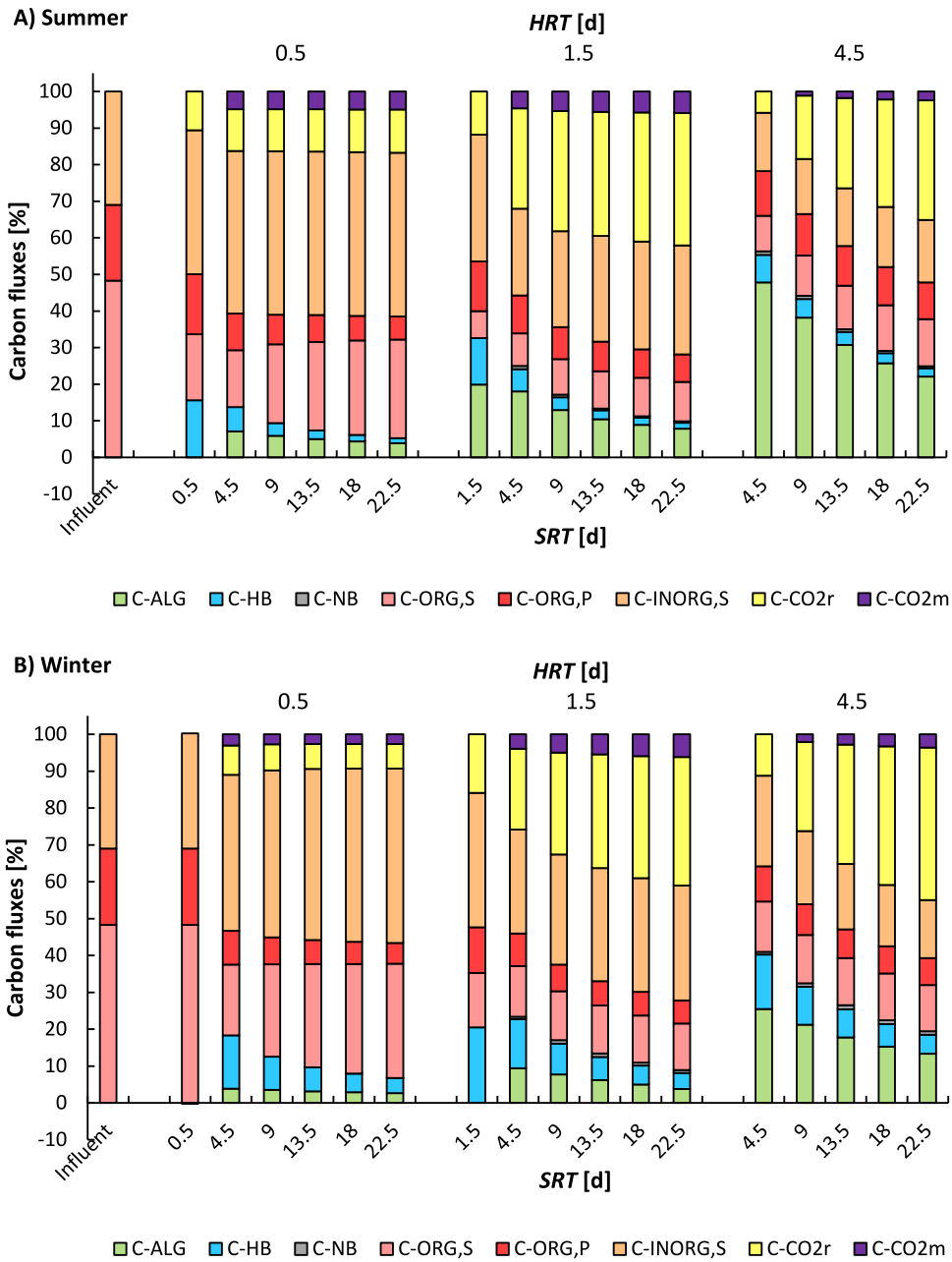


**Fig. 5.** Carbon flux percentages of the influent and effluent under different scenarios of SRT (1.5, 4.5, 9,13.5, 18, 22.5 [d]) and HRT (0.5, 1.5, 4.5 [d]), for summer (A) and winter (B) conditions. C-ORG,P corresponds to the carbon presented in $X_S$, $X_I$; C-ORG,S corresponds to the soluble carbon fraction in $S_s$ and $S_I$; C-INORG,S is the soluble inorganic carbon modelled as $S_{IC}$; C-ALG, C-NB, and C-HB correspond to the fraction of carbon into the algal, NB (AOB and NOB) and HB biomass, respectively. C-CO2r and C-CO2m are the inorganic carbon fraction in the form of $CO_2$ gas, in the reactor and the membrane compartments, respectively.

$CO_2$ partial pressures in both compartments (i.e. $S^*_{CO2,R}$ and $S^*_{CO2,m}$):

$$\frac{CTRr}{CTRm} = \frac{K_L a_r \left(S^*_{CO2,R} - S_{CO2,R}\right) V_r}{K_L a_m \left(S^*_{CO2,m} - S_{CO2,m}\right) V_m} \tag{10}$$

If we assume similar concentrations in the reactor and in the membrane modules, then the contribution of the membrane tank to $CO_2$ stripping is determined by the ratio of $K_La$ and volumes of each compartment. If the volume ratio largely exceeds the ratio of $K_La$, then the membrane tank should not contribute largely to the desorption of $CO_2$. This is true for the system in most of the examined cases. For instance, when HRT = 2.5, 3.5 or 4.5 [d], only about 12 % of $CO_2$ desorption would be attributed to membrane tank. This conclusion is also applicable to the mass transfer of other gases such as oxygen, as it will be discussed below.

Direct $CO_2$ emissions from wastewater treatment plants are conventionally excluded from the greenhouse gas emission inventory, as proposed by the International Panel on Climate Change (Bartram et al., 2019). However, some authors have highlighted the importance to consider such emissions, since some of them are coming from the decomposition of organic matter present in carbon-based fossil pollutants, like detergents, cosmetics, and pharmaceuticals (Bao et al., 2015). Monteith et al. (2005) estimated that the on-site direct $CO_2$ emissions associated with removal of organic matter for Canadian conventional wastewater treatment plants was around 1.4 g $CO_2$ per g COD removed which is superior to the estimated emissions of membrane assisted MBC that was lower than 1.27 g $CO_2$ g $COD^{-1}$, depending on the applied condition.

### 3.2.3. Nitrogen mass balances

The distribution of the nitrogen fluxes as predicted by M-ALBA is shown in Fig. 6. As for carbon balance, the nitrogen fluxes are grouped in the following forms:

- N-ORG: Nitrogen in organic matter other than biomass (i.e. $X_S$, $S_S$, $X_I$, and $S_I$) and soluble nitrogen ($S_{ND}$).
- N-ALG: Nitrogen in algae; N-HB: Nitrogen in heterotrophic bacteria; and N-NB: Nitrogen in nitrifying bacteria.
- N-TAN: Total ammoniacal nitrogen, $S_{NH}$.
- N-$NO_2$ and N-$NO_3$: Nitrous nitrogen and nitric nitrogen, respectively.
- N-$NH_3$r and N-$NH_3$m: Ammonia gas, in the reactor and the membrane compartments, respectively.
- N-$N_2$: Nitrogen gas.

Contribution of nitrogen fluxes associated with biomass exhibit similar trends as those for carbon, for the explored ranges of SRT and HRT: at higher SRTs the nitrogen fluxes associated to microorganisms are lower. Notably, the maximum contribution of algal biomass to nitrogen outflow was <20 %, at HRT 4.5 [d] for the summer scenario.

During summer, the consumption of organic nitrogen N-ORG (i.e. urea hydrolysis or ammonification and the conversion of slow-biodegradable COD into easily biodegradable) is strongly enhanced at higher temperatures. This appears clearly in Fig. 6, when comparing summer and winter scenarios. For instance, at HRT = 4.5 [d] and SRT = 13.5 [d], the predicted ammonification rates were approximately 1.5 times higher in summer than in winter, despite the population of heterotrophs were much lower in summer. Higher ammonification rates lead to an increase in ammoniacal nitrogen (N-TAN) available for

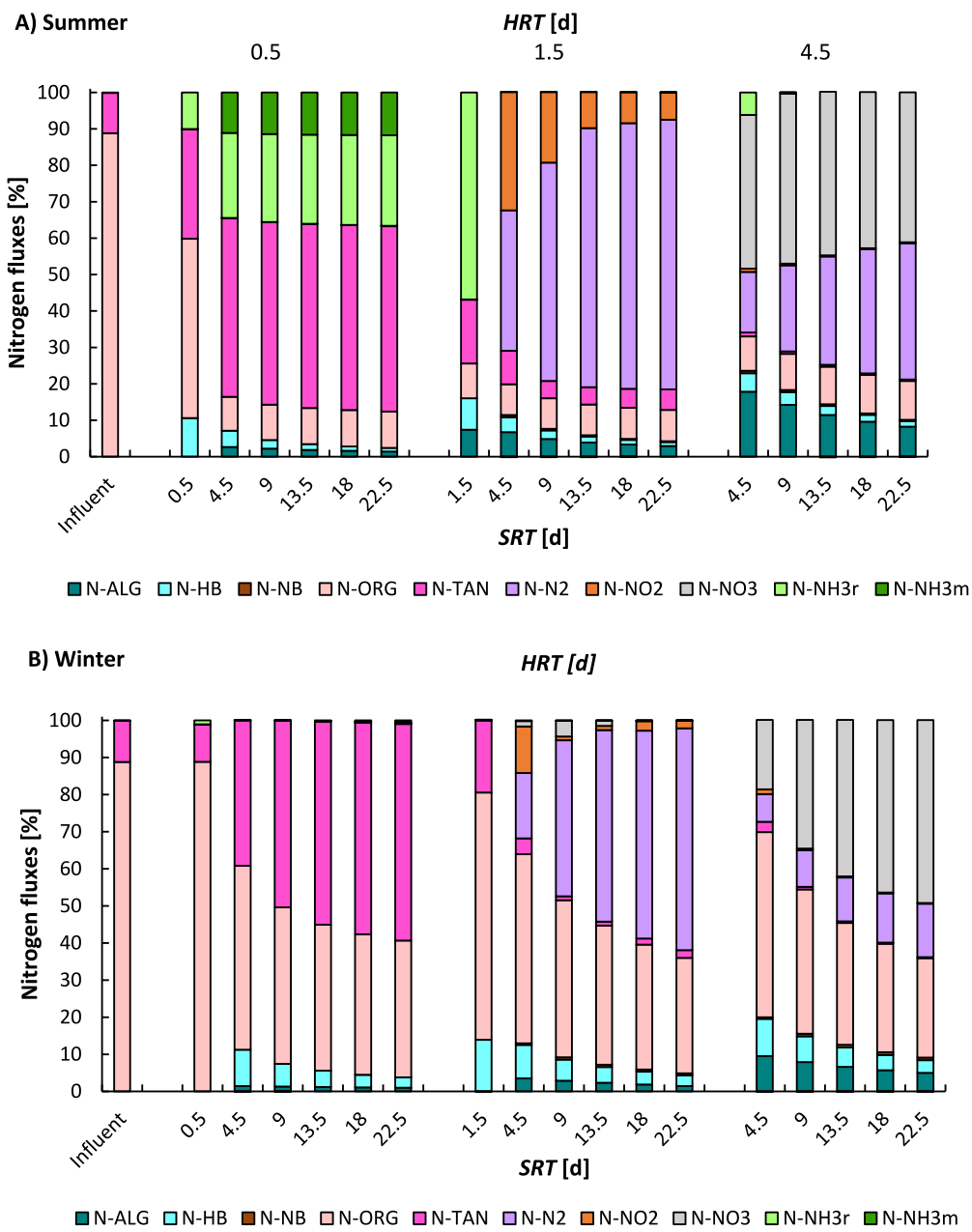


**Fig. 6.** Nitrogen fluxes as percentage of the influent and effluent for different scenarios of SRT (1.5, 4.5, 9,13.5, 18, 22.5 [d]) and HRT (0.5, 1.5, 4.5 [d]), for Summer (A) and Winter (B) conditions. N-ORG is the organic nitrogen in $X_s$, $S_S$, $X_I$, $X_s$, and as organic nitrogen (modelled as variable $S_{ND}$). N-ALG, N-HB, and N-NB are the fraction of algal, NB (AOB and NOB), and HB biomass, respectively. N-TAN is the ammoniacal nitrogen fraction ($S_{NH}$), and N-$NO_2^-$ y N-$NO_3^-$ are the fraction of nitrogen as nitrite and nitrate, respectively. N-$NH_3$r and N-$NH_3$m are the fluxes of ammonia gas, in the reactor and the membrane compartments, respectively. N–$N_2$ is the flux of nitrogen gas.

bacteria and algae.

As observed in Fig. 6, the ammonia stripping, N-$NH_3$r and N-$NH_3$m, occurs mainly during summer scenarios, and only when HRT = 0.5 [d] or SRT equal to HRT. pH was not relevant for ammonia stripping since the values barely exceeded 9. When SRT ≥ 4.5, HRT = 0.5 [d], in summer, about 35 % of the incoming nitrogen (24 % as N-$NH_3$r and 11 % as N-$NH_3$m) was released into the atmosphere as ammonia and around 50 % as N-TAN. In contrast, in all the other scenarios when *HRT* > 0.5 [d] and SRT > HRT, NB was present in the system and N-TAN constituted <10 % (below 1 % in most cases) of the produced nitrogen, preventing the stripping of ammonia. This is a positive outcome, since atmospheric $NH_3$ is very harmful for the environment, playing a significant role in acid rains, nucleating fine particulates, and eutrophicating sensitive ecosystems (Behera et al., 2013).

M-ALBA predicts active denitrification at HRT = 1.5 [d], with a further enhancement when increasing SRT. According to Fig. 6, at low HRT (0.5 [d]), the washout of nitrifiers resulted in the absence of nitrate or nitrite, while at higher HRT (4.5 [d]), complete nitrification occurs, with large concentrations of N-$NO_3$ and the absence of N-$NO_2$. At an HRT of 1.5 [d], the predicted presence of nitrite revealed partial nitrification, resulting from oxygen limitation, as discussed in the next section. However, a great part of nitrogen was converted into nitrogen gas N–$N_2$, especially when increasing SRTs, given the limiting dissolved oxygen concentration in these scenarios. N–$N_2$ ranged from 39 to 74 % of the total outflow of nitrogen for summer scenarios and from 18 to 60 % for winter.

Model predicts that under many of the tested conditions relevant amounts of nitrate will be present in the effluent, so additional stages or

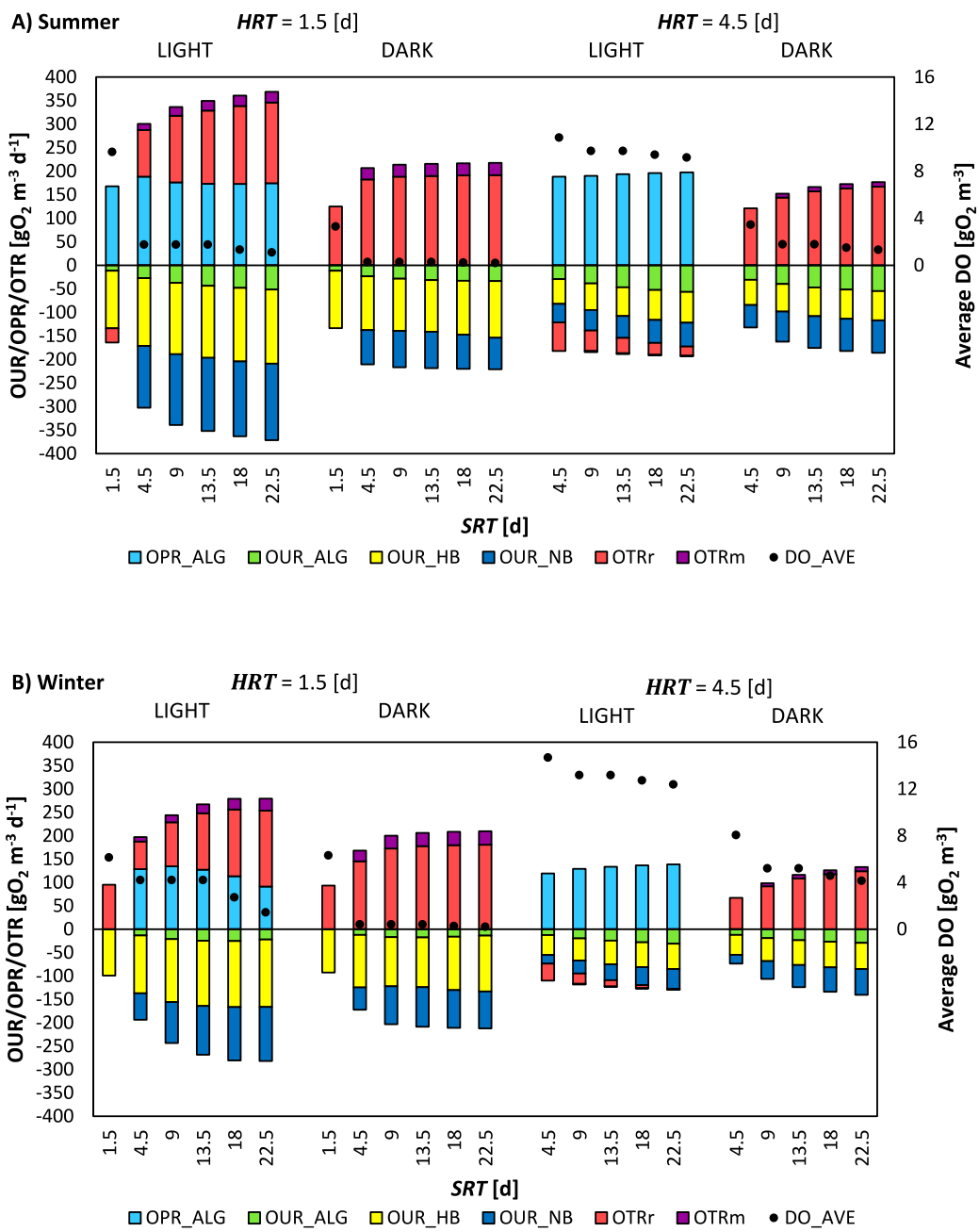


**Fig. 7.** Oxygen uptake rate (OUR), oxygen production rate (OPR) and oxygen transfer rate (OTR) under different scenarios of SRT (1.5, 4.5, 9,13.5, 18, 22.5 [d]) and HRT (0.5, 1.5, 2.5, 3.4, 4.5 [d]), for summer (A) and winter (B) conditions. OPR_ALG is the oxygen production rate of algae; OUR_ALG, OUR_HB, y OUR_NB are the oxygen uptake rate of algae, heterotrophs, and nitrifiers, respectively. OTRr and OTRm are the transference rate of oxygen of the reactor and the membrane compartments, respectively. Black dots indicate the dissolved oxygen (DO_AVE) in the liquid phase.

processes for its removal may be needed. Notably, several studies reported the addition of an external carbon source during the night, as an efficient way to enhance the denitrification process rate in MBC wastewater treatment systems (Arun et al., 2019; Liu et al., 2022; Wang et al., 2015).

### 3.2.4. Effect of HRT and SRT on dissolved oxygen dynamics

Dissolved oxygen concentration is a crucial factor for MBC since it supports aerobic bacterial growth and nitrification. Fig. 7 presents the oxygen fluxes, that were evaluated with the following considerations:

- OUR_ALG, OUR_HB and OUR_NB: Oxygen uptake rate of microalgae, heterotrophs and nitrifiers, respectively. Since it is a consumption rate, it appears as negative values.
- OPR_ALG: Oxygen production rate of algae during photosynthesis, with positive values.
- OTRr and OTRm: Gas-liquid oxygen transfer rate from gas to liquid, in the reactor and the membrane compartments, respectively. These values can be negative or positive, depending on oxygen desorption and absorption dynamics.

In addition, a distinction was made between light and dark phases, and the predicted average oxygen concentrations (DO_AVE) are also reported (black dots).

As expected, higher biological oxygen demands, OUR_ALG, OUR_HB and OUR_NB, resulted from higher biomass concentrations due to its retention (see Fig. 4 and Fig. 7). Despite the small proportion of nitrifiers, the associated oxygen consumption was marked (OUR_NB).

Environmental conditions, such as temperature and irradiance, are expected to play a key role on oxygen dynamics, as confirmed by the simulations presented in Fig. 7. Oxygen uptake rate was higher during summer, especially due to microalgae and nitrifiers. In fact, warmer scenarios involve higher bacterial and algal respiration rates, thus increasing oxygen demand. An increase in temperature also decreases oxygen solubility, affecting gas/liquid mass transfer. Indeed, the estimated average oxygen saturation in summer was 30 % lower than in winter. Fig. 7 shows that during the light phase, for HRT = 4.5 [d], microalgae were able to fully meet the oxygen demand of bacteria. However, in the dark phase, the conditions were completely different, and DO levels dropped down drastically, leading to oxygen limitation.

As already discussed, the importance of the membrane tank on gas/liquid mass transfer depends on the relation between $K_La$ values and the system compartment volumes (reactor and membrane). In the standard configuration where the membrane compartment is much smaller than the reactor, the gaseous transfer in the membrane compartment is much lower than in the reactor. This can be seen clearly in Fig. 7 for the case of oxygen dynamics. Simulations revealed the importance of the combined OTR (OTRr + OTRm) on the oxygen dynamics, especially at low HRTs (when oxygen requirements are higher), and during the dark phases when photosynthesis does not occur. For example, during summer at HRT = 1.5 [d] and SRT = 22.5 [d], OTR was similar to the oxygen production by algae (OPR_ALG). However, OTR was higher during winter, since oxygen saturation is higher at lower temperatures. During the dark phase, OTR was the only supply of oxygen, supporting a moderated activity of heterotrophs and nitrifiers. Over the full cycle of 24 h, OTR provided 2/3 of the oxygen for HRT = 1.5 [d] and 1/2 for HRT = 4.5 [d], while algae provided the remaining fraction.

### 3.2.5. Efficiency of the wastewater remediation system

The M-ALBA model was also used to predict the COD and nitrogen concentrations in the effluent, as indicators of the remediation system efficiency. The average effluent (permeate) concentrations of total soluble nitrogen and COD are shown in Figs. 8 and 9. The discharge limits for treated domestic wastewater imposed by the European directive 91/271/EEC are presented for comparison purposes. The scenarios where SRT = HRT, i.e. without the membrane module, are presented to highlight the advantages of the membrane separation system.

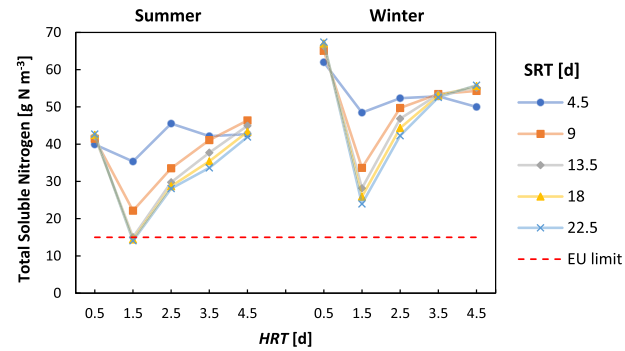


**Fig. 8.** Soluble nitrogen concentration in the effluent different scenarios of SRT (1.5, 4.5, 9,13.5, 18, 22.5 [d]) and HRT (0.5, 1.5, 2.5, 3.4, 4.5 [d]), for Summer (A) and Winter (B) conditions. The dashed lines represent the discharge limit concentration in European directive 91/271/EEC (15 [g N $m^{-3}$]).

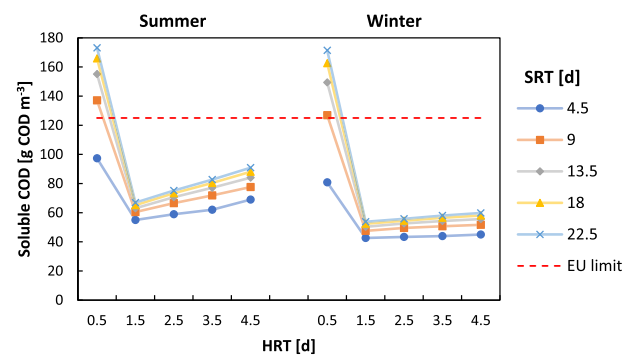


**Fig. 9.** Soluble COD concentration ($S_S$ and $S_I$) in the effluent for the different scenarios: SRT (1.5, 4.5, 9,13.5, 18, 22.5 [d]) and HRT (1.5, 4.5 [d]), for summer and winter conditions. The dashed lines represent the discharge limit concentration in European directive 91/271/EEC (125 [g COD $m^{-3}$]).

Fig. 8 shows that the nitrogen content in the effluent decreased with longer SRT and shorter *HRT*. Fig. 8 reveals a minimum of total soluble nitrogen concentration for HRT around 1.5 [d] and SRT >9 [d]. For this condition, the overall nitrogen removal was in average 61.5 % for winter and 78.5 % for summer. Summer scenarios exhibited better results, thanks to the higher temperature, allowing larger removal rate capacity. Improvement of nitrogen removal for lower HRTs is the result of oxygen limitation at higher biomasses. Such conditions are typically favourable for denitrification. Oxygen limitation during the night can be a desired operational mode, to promote denitrification, enhancing overall nitrogen removal.

Even though simulations do not predict nitrogen level coping with the selected discharge standard, it is observed that adding a membrane to decouple SRT from HRT increases the water flow that can be effectively treated. This allows operation at lower HRT values and surface requirement than are commonly required (Chen et al., 2003; Kim et al., 2014; Solimeno and García, 2019), without compromising the effluent quality.

The effluent soluble COD (accounted as the sum of $S_{S,m}$ and $S_{I,m}$ in the permeate) is presented in Fig. 9 for all the analysed scenarios. The model predicts concentrations below the European discharge standard values (125 [g COD $m^{-3}$]) for all the SRT and HRT, except at HRT = 0.5 [d], where the influent flow exceeds the system's treatment capacity. Soluble COD removal efficiencies were higher in winter scenarios (over 75 %) compared to summer scenarios (above 62 %), because of the negative impact of higher temperatures on HB growth rates. Finally, as for nitrogen removal, optimal COD removal was obtained for HRT of at least 1.5 [d].

### 3.3. Discussion

Mathematical modelling and simulation are useful tools to provide insights about the behaviour of complex systems, when responses to operational conditions are difficult to predict due to the nonlinear interactions within the players of the ecosystem, submitted to permanent light and temperature fluctuations. Exploring the behaviour and performances of this system under a variety of scenarios revealed the benefits of adding a membrane-assisted biomass retention step. Results show how different levels of biomass retention and wastewater load (determined by HRT) must be optimized and adapted to the climatic conditions to maintain efficient performance.

Even though hollow fibre membranes require intense aeration to reduce biofouling, the reduced relative volume of the membrane tank should not significantly impact the gas/liquid mass transfer.

The biomass retention triggered by the membrane has been demonstrated to deeply impact the carbon dynamics and distribution. It is generally claimed that the advantage of using microalgae is the fixation of anthropogenic $CO_2$. However, as a consequence of biomass retention, the carbon dioxide production increases, and the access to light for phototrophic organisms decreases, resulting in systems that release more carbon dioxide than what they absorb. More refined computations, involving Life Cycle Assessment (Morales et al., 2019), should therefore be carried out to better quantify the gain in carbon footprint compared to the traditional processes. Simulations also show how, despite the contribution of algae to oxygen dynamics, absorption from atmosphere is still a relevant source of oxygen, because of the large area of HRABP systems. The M-ALBA model predicted oxygen concentrations in the medium that under certain conditions can peak up above 20 $mgO_2/L$ during the day, potentially triggering strong oxidative stress, driving molecular degradation (Matamoros et al., 2015). This specificity of microalgae-based process is probably understated yet and would need more work to be better appreciated.

Simulations point out that a substantial reduction on HRT has a marginal impact on the systems performance in terms of organic matter removal. Therefore, since SRT and HRT moderately impact the removal of organic matter, it is recommended to determine the operating conditions, and especially HRT, from the targeted performances for nitrogen removal and from the installation size. In a second step and to optimize the process performance, operational settings for SRT must be chosen to exceed HRT. Moreover, to mitigate ammonia volatilization, low HRT are desired for enhancing nitrogen removal in the form of $N_2$. Specifically, in the tested climatology, the process can be optimally operated with an HRT of approximately 1.5 [d] and an SRT greater than or equal to 9 [d]. This strategy turns out to significantly increase the process performance. The wastewater treatment capacity when operated at an HRT of 1.5 [d] is over 4-fold the treatment capacity with HRT = 7 [d], condition that can be considered a conservative operational condition for HRABP without biomass retention. In other words, following these recommendations, the HRABP area requirement per person equivalent would decrease over 4 times.

## 4. Conclusions

The M-ALBA model was proposed as an extension of the validated ALBA model to represent biomass retention in HRABP systems coupled to membrane filtration. Experimental data were used to successfully validate the model prediction capability. Model predictions demonstrated the effectiveness of uncoupling SRT from HRT in a MBC based system, providing the following main outcomes:

- Biomass retention increases biomass respiration rate, which results in lower net biomass productivity and the release of more biogenic $CO_2$.
- Retention of slow growing nitrifying bacteria enables total nitrification, therefore reducing consistently ammonia stripping.
- Biomass retention allows operation at low HRT, which increase oxygen requirements. Under certain conditions, such limitation during the night promotes denitrification, favouring the overall nitrogen removal.
- Even though intense mixing and air sparging is expected in the membrane compartment for fouling control of the membrane, the contribution of this compartment to overall gas-liquid mass transfer stays moderated.
- The increase of the treatment capacity resulting from biomass retention, enables the reduction of the area and volume required. This may contribute to significantly reduce the resources and cost requirements, favouring real-scale application.

Based on these results obtained from extensive simulations, adding a filtration unit for biomass recirculation to the reactor represents a significant opportunity to quantitative improve domestic wastewater treatment, particularly in terms of nitrogen compound removal and, more significantly, volumetric treatment capacity.

## CRediT authorship contribution statement

**François Crouchett-Catalán:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Jineth Arango:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Olivier Bernard:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Carlos Martínez:** Writing – review & editing, Writing – original draft, Validation. **Francesca Casagli:** Writing – review & editing, Writing – original draft, Validation, Supervision, Formal analysis, Conceptualization. **David Jeison:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: David Jeison reports financial support was provided by National Agency for Research and Innovation. Francesca Casagli and Olivier Bernard reports financial support was provided by The French Agency for Ecological Transition. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The authors are unable or have chosen not to specify which data has been used.

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